Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/07/2023
PI Name:	Wargo, Matthew Ph.D.		
Project Title:	Impact of Simulated Microgravity on Potable Water Bacterial Biofilm Formation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	05405	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR-Space Biology NNH20ZHA001C
Start Date:	07/01/2020	**End Date:**	06/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA EPSCoR
Contact Monitor:	Compton, Jeppie	**Contact Phone:**	
Contact Email:	jeppie.r.compton@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2022 (original end date was 6/30/2021) per NSSC information (Ed., 1/18/22)		
Key Personnel Changes/Previous PI:	Dr. Bernard Cole serves as the Vermont NASA EPSCoR Director, he is the administrative Principal Investigator on this proposal and will oversee the overall progress of the work and report to NASA.		
COI Name (Institution):	Lee, Jessica Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	80NSSC20M0158		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Established Program to Stimulate Competitive Research (EPSCoR) Project All human habitation in space will take place in a non-sterile environment and the multispecies bacterial biofilms in the drinking water are an important source of bacteria to which all astronauts are exposed. This constant exposure to potential bacterial pathogens and the additional roles of biofilms in material corrosion and chemical processes make it imperative that we improve our understanding of these co-habitants on our space-going vessels. This proposal addresses the Biology of the Built Environment topic, under the Space Biology Program within the Space Life and Physical Sciences Research and Applications (SLPSRA) division [Ed. Note: as of August 2020, it is Biological and Physical Sciences Division]. The call notes a particular interest in relation to this topic on determining the impact of simulated microgravity on mixed microbial biofilms and community interactions. Our NASA-related work has focused on studying community interactions driving biofilm formation and physiology by members of the bacterial biofilm present in the Potable Water Reclamation System of the International Space Station (ISS). These species are a model of the ISS community and also model drinking water communities in general, including those destined for future space flights, as these species are common globally in municipal water systems. During the course of our previous NASA Established Program to Stimulate Competitive Research (EPSCoR)-funded research, a six-member bacterial community derived from the ISS potable water reclamation system was characterized and used as a model community to understand biofilm assembly in potable water. This work led to three major findings relevant to this proposal: (i) ISS water system biofilm formation is robust to community composition, (ii) Determination of the species interaction network governing biofilm assembly, and (iii) Species interactions suppress pathogen survival. This proposal will follow up on the initial findings to achieve two goals: (1) Measure the impact of low-shear simulated microgravity on the community interactions and biofilm formation network in the model ISS biofilm community. (2) Determine if the ability of ISS biofilm members to limit bacterial pathogen growth and invasion of the community continues in low shear simulated microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Final summary of project progress per Dr. Wargo: For Objective 1, we first tested the biofilm community assembly model that arose from our previous work – that Sphingomonas paucimobilis and Chryseobacterium gleum were the key nodes in the biofilm formation network and that C. gleum's role requires interaction with either Cupriavidus metallidurans or Ralstonia insidiosa. We found that the community assembly rules still hold for the RWV low shear environment. Total bacterial growth is not significantly different between these three groups but the full community mix does form substantial aggregates. Additionally, we examined the biofilm enhancement between C. gleum and R. insidiosa, which is a key interaction in the assembly model. For that interaction, we saw that while the co-culture does lead to enhanced biofilm formation and this is part of the pre-print currently posted with the manuscript under review at Microbiology Spectrum. [Ed. Note: See Bibliography.] For Objective 2, we tested the ability of the multispecies community to suppress the growth of the opportunistic pathogens Pseudomonas aeruginosa, Stenotrophomonas maltophilia, and Klebsiella pneumoniae. For P. aeruginosa, culture under static conditions demonstrated suppression of by the full species mix or R. insidiosa alone. In the Rotating Wall Vessel (RWV) system, the full species mix suppressed P. aeruginosa growth and R. insidiosa was still inhibitory. For S. maltophilia, static cultures showed suppression by the full species mix and S. paucimobilis. In the RWV conditions, both the full species mix and S. paucimobilis were able to suppress S. maltophilia growth. This strain of K. pneumoniae showed tremendous capsule (polysaccharide) production during RWV growth, so much so that the full volume became effectively a gel. Because of this, there was no suppression of K. pneumonia growth and we did not continue with this portion of the sub-aim. Importantly, the K. pneumonia strain from the International Space Station (ISS) produces much lass capsule under these conditions, so the soon-to-be-started study with Jet Propulsion Laboratory (JPL) can continue. The work on the pathogen co-culture is being incorporated into static culture work ongoing in the lab. as part of a manuscript in preparation. Key outcomes: The biofilm assembly rules for the model ISS potable water community, first determined in a static system at terrestrial gravity conditions, do still hold for the RWV system which mimics the low-shear present under microgravity conditions. The suppression of pathogens by the ISS model community and select individual members holds for P. aeruginosa and S. maltophilia. Capsule formation by K. pnuemoniae in the RWV system prevent substantial competition and make the experiment technically uninformative. (Ed., 4/4/23)
Bibliography Type:	Description: (Last Updated: 12/21/2023)
Articles in Peer-reviewed Journals	Foote A, Schutz K, Zhao Z, DiGianivittorio P, Korwin-Mihavics BR, LiPuma JJ, Wargo MJ. "Characterizing biofilm interactions between Ralstonia insidiosa and Chryseobacterium gleum." Microbiology Spectrum. 2023 Apr 13;11(2):e04105-22. https://doi.org/10.1128/spectrum.04105-22 , Apr-2023